

# Correlation

## WHAT IS CORRELATION?

**Correlation** is a statistical technique that tells us **how strongly two variables are related**.

It answers questions like:

- If one value increases, does the other also increase?
- If one value increases, does the other decrease?
- Is there any relationship between them?

### Simple Definition

**Correlation measures the strength and direction of the relationship between two variables.**

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## REAL-LIFE EXAMPLE 1

Suppose you study every day.

### Hours Studied Marks

|   |    |
|---|----|
| 1 | 40 |
| 2 | 50 |
| 3 | 60 |
| 4 | 70 |
| 5 | 80 |

As study hours increase, marks also increase.

This is called **Positive Correlation**.

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## REAL-LIFE EXAMPLE 2

### Speed Time Taken

|    |    |
|----|----|
| 20 | 10 |
| 40 | 5  |
| 60 | 3  |
| 80 | 2  |

As speed increases, time decreases.

This is **Negative Correlation**.

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### REAL-LIFE EXAMPLE 3

#### Shoe Size IQ

|    |     |
|----|-----|
| 7  | 90  |
| 8  | 110 |
| 9  | 95  |
| 10 | 105 |

There is no relationship.

This is **No Correlation**.

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## TYPES OF CORRELATION

### 1. Positive Correlation (+)

Both variables move in the **same direction**.

Example

Study Hours ↑  
Marks ↑

Another example

Experience ↑  
Salary ↑

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### 2. Negative Correlation (-)

Variables move in **opposite directions**.

Example

Temperature ↑  
Jacket Sales ↓

Another example

Speed ↑  
Travel Time ↓

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### 3. Zero Correlation

No relationship exists.

Example

Hair Length

Exam Marks

One does not affect the other.

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## CORRELATION VALUE

Correlation is always between

-1 and +1

### Correlation Meaning

|      |                   |
|------|-------------------|
| +1   | Perfect Positive  |
| +0.8 | Strong Positive   |
| +0.5 | Moderate Positive |
| 0    | No Relationship   |
| -0.5 | Moderate Negative |
| -0.8 | Strong Negative   |
| -1   | Perfect Negative  |

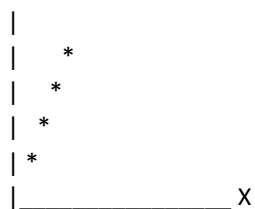
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## GRAPHICAL REPRESENTATION

### Positive Correlation

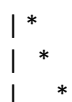
Y

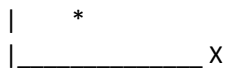


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### Negative Correlation

Y





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### No Correlation

Y



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### Why Do We Use Correlation?

Suppose we have data

#### Study Hours Attendance Marks

|   |    |    |
|---|----|----|
| 2 | 80 | 40 |
| 3 | 85 | 50 |
| 5 | 90 | 70 |

We want to know

Which feature affects Marks most?

Correlation tells us.

Maybe

Study Hours → 0.95

Attendance → 0.45

So Study Hours is more important.

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### WHY IS CORRELATION IMPORTANT IN MACHINE LEARNING?

Correlation is used during **Exploratory Data Analysis (EDA)** and **Feature Selection**.

It helps us:

- Understand relationships between variables
- Select useful input features
- Remove duplicate information
- Improve model performance
- Reduce overfitting

- Simplify the model

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### EXAMPLE IN MACHINE LEARNING

Suppose we want to predict **House Price**.

Dataset

#### Area Bedrooms Price

1000 2 30

1200 2 35

1500 3 45

1800 3 55

2200 4 70

Correlation with Price

#### Feature Correlation

Area 0.96

Bedrooms 0.72

Conclusion

Area is more useful than Bedrooms.

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### Another Example

Suppose we predict Student Marks.

Features

- Study Hours
- Mobile Usage
- Attendance

Correlation

#### Feature Correlation with Marks

Study Hours 0.91

Attendance 0.70

| Feature | Correlation with Marks |
|---------|------------------------|
|---------|------------------------|

Mobile Usage -0.82

Interpretation

Study Hours increase → Marks increase.

Attendance increase → Marks increase.

Mobile Usage increase → Marks decrease.

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## WHERE IS CORRELATION USED IN ML?

### 1. Feature Selection

Remove weak features.

Example

Height → 0.85

Blood Group → 0.02

Blood Group can be removed.

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### 2. Remove Duplicate Features (Multicollinearity)

Example

|Area (sq.ft)|Area (sq.meter)|

Both represent the same information.

Correlation

0.99

Keep only one feature.

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### 3. Exploratory Data Analysis

Understand data before training.

Questions answered:

- Which feature is important?
- Which feature has no effect?
- Which features are highly related?

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### 4. Data Visualization

Use Heatmap to quickly understand feature relationships.

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### Correlation Formula (Pearson Correlation)

The most common correlation coefficient is:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Where:

- $x_i$  = values of X
- $y_i$  = values of Y
- $\bar{x}$  = mean of X
- $\bar{y}$  = mean of Y
- $r$  = correlation coefficient

The formula compares how X and Y vary together (covariance) and scales it by their standard deviations, producing a value between -1 and +1.

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### Python Example 1

```
import pandas as pd
```

```
data = {  
    "Study_Hours": [2, 3, 4, 5, 6],  
    "Marks": [40, 50, 60, 70, 80]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df.corr())
```

Output

```
      Study_Hours  Marks  
Study_Hours    1.00    1.00  
Marks          1.00    1.00
```

Perfect positive correlation.

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### Python Example 2

```
import pandas as pd
```

```
data = {  
    "Mobile_Usage": [8, 7, 6, 5, 4],  
    "Marks": [40, 50, 60, 70, 80]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df.corr())
```

Output

|  | Mobile_Usage | Marks |
|--|--------------|-------|
|--|--------------|-------|

|              |       |       |
|--------------|-------|-------|
| Mobile_Usage | 1.00  | -1.00 |
| Marks        | -1.00 | 1.00  |

Perfect negative correlation.

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### Python Example 3

```
import pandas as pd
```

```
data = {  
    "Study_Hours":[2,4,6,8,10],  
    "Attendance":[60,70,80,90,95],  
    "Marks":[30,45,60,75,90]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df.corr())
```

Possible Output

|  | Study_Hours | Attendance | Marks |
|--|-------------|------------|-------|
|--|-------------|------------|-------|

|             |      |      |      |
|-------------|------|------|------|
| Study_Hours | 1.00 | 0.96 | 0.99 |
| Attendance  | 0.96 | 1.00 | 0.95 |
| Marks       | 0.99 | 0.95 | 1.00 |

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### Correlation Heatmap

```
import pandas as pd
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
data = {  
    "Study_Hours":[2,4,6,8,10],  
    "Attendance":[60,70,80,90,95],  
    "Marks":[30,45,60,75,90]  
}
```

```
df = pd.DataFrame(data)
```

```
sns.heatmap(df.corr(),  
            annot=True,  
            cmap="coolwarm")
```



```
plt.show()
```

#### Explanation

- `df.corr()` calculates the correlation matrix.
  - `annot=True` displays the correlation values in each cell.
  - `cmap="coolwarm"` uses blue for negative correlations and red for positive correlations (with lighter colors near zero).
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#### Different Correlation Methods in Pandas

```
df.corr(method="pearson")
```

- Most common.
- Measures **linear** relationships.
- Used when data is continuous and approximately normally distributed.

```
df.corr(method="spearman")
```

- Based on **rank** rather than actual values.
- Useful for monotonic relationships or when data contains outliers.

```
df.corr(method="kendall")
```

- Also rank-based.
  - Often used with small datasets or when there are many tied ranks.
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#### Advantages

- Easy to calculate
  - Easy to understand
  - Helps select important features
  - Improves model performance
  - Detects redundant variables
  - Useful during EDA
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#### Limitations

- Correlation **does not imply causation**. Two variables can be highly correlated without one causing the other.
- Pearson correlation captures only **linear** relationships. A strong non-linear relationship may still produce a low Pearson correlation.
- Sensitive to outliers, which can distort the correlation value.

- A correlation near zero does not always mean there is no relationship—it may simply be non-linear.